Synopsis

On

Bhunaga: A Multilingual Retrieval-Augmented Advisory System for Indian Farmers

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1. Title of the Project

Bhunaga: A Multilingual Retrieval-Augmented Advisory System for Indian Farmers

2. Abstract

Farmers across India often think and communicate in their regional language, yet the agricultural information they need most — crop advisories, weather forecasts, pest-control guidance and mandi (market) prices — is scattered across many websites and largely available only in English. As a result, many farmers depend on local pesticide retailers for advice, which can lead to unnecessary chemical use and higher costs, or on generic AI chatbots, which may produce agriculturally incorrect recommendations. Bhunaga is a multilingual, retrieval-augmented advisory system designed to close this gap. Farmers ask questions about sowing, pest control, weather, mandi prices and selling produce through voice or text, in their own language, via an app, WhatsApp or an IVR call. The system identifies the language and intent, retrieves relevant information from verified agricultural knowledge bases and live APIs (weather, mandi prices, government schemes), grounds its answer in these sources through a retrieval-augmented generation pipeline with a trust-and-validation check, and delivers a localised, source-cited response back to the farmer in their own language. The proposed system aims to provide an affordable, trustworthy and easy-to-use alternative to scattered, English-only agricultural information sources.

3. Introduction / Background

India is home to a very large number of farming households, most of whom conduct their daily lives and decision-making in a regional language rather than English. Sowing decisions, pest-control choices, and the timing of selling produce all depend on timely, accurate information — yet weather bulletins, crop-advisory material and mandi price boards are usually published in English and spread across many separate government and private websites. This gap in access pushes many farmers to rely on local pesticide retailers, whose advice is often driven by what they stock rather than what the field actually needs, resulting in overuse of chemicals and avoidable cost.

The recent growth of large language models (LLMs) and retrieval-augmented generation (RAG) has made it possible to build systems that understand natural-language questions and produce grounded, source-backed answers rather than free-form guesses. However, general-purpose AI chatbots are not tuned for regional languages or Indian agricultural conditions, and can generate confident-sounding but agriculturally incorrect recommendations. This project, Bhunaga, combines multilingual natural-language processing with a retrieval-augmented pipeline over verified agricultural sources and live data feeds, so that farmers receive advice that is both understandable in their own language and grounded in trustworthy information.

4. Problem Definition

Farmers in India often make farming decisions in their regional language, but reliable agricultural information — such as crop advisories, weather updates and mandi prices — is mostly available in English and spread across multiple websites. As a result, farmers struggle to access timely and accurate guidance on sowing, pest control and spraying, and selling their produce. Many depend on local pesticide retailers for advice, which can lead to unnecessary chemical use and higher cost. Existing apps provide only partial information, while general AI chatbots may generate incorrect agricultural recommendations. The problem, therefore, is the lack of a trusted, easy-to-use system that delivers accurate, localised and timely farming advice directly in the farmer's own language.

5. Objectives

1. To accept farmer queries as voice or text, in a regional language, through an app, WhatsApp, or an IVR call.

2. To identify the query language and translate it for downstream processing, and to classify the farmer's intent (sowing, pest control, weather, mandi price, or selling produce).

3. To retrieve relevant information from verified agricultural knowledge bases and live data sources (weather API, mandi price API, government scheme database).

4. To generate a grounded, source-cited answer using a retrieval-augmented generation (RAG) pipeline, with a validation step to reduce incorrect or unsupported claims.

5. To translate the generated answer back into the farmer's own language and deliver it through the same channel used for the query.

6. To reduce farmers' dependence on local pesticide retailers and generic chatbots for agricultural decisions.

6. Scope of the Project

The scope of this project covers advisory queries relating to sowing guidance, pest control and spraying, weather forecasts, mandi (market) prices, and selling/marketplace information, delivered in major Indian regional languages through voice or text. The system focuses on retrieval-augmented, source-grounded advice rather than autonomous decision-making — final decisions remain with the farmer. It does not include physical automation, on-field sensing hardware, or financial transaction processing; these are considered possible future extensions. The initial version is scoped to a limited set of regional languages and a defined set of verified agricultural knowledge sources.

7. Proposed System

The proposed system, Bhunaga, is built around a retrieval-augmented generation (RAG) pipeline. A farmer's query, given as voice or text in their own language through an app, WhatsApp, or an IVR call, first passes through a Language and Preprocessing layer, which identifies the language, converts speech to text where needed, and translates the query into English for processing. An Intent Classifier then determines whether the query relates to sowing guidance, pest control and spraying, weather, mandi price, selling/marketplace, or government schemes.

Based on the identified intent, a Retrieval Layer searches a combination of vector-indexed knowledge bases (verified crop-advisory and pest-control documents, government scheme details) and live APIs (real-time weather forecasts, mandi price feeds, marketplace/buyer information) to gather relevant context. A Trust and Validation layer cross-checks the draft answer against this retrieved context and flags unsupported claims, reducing the risk of the hallucinated or agriculturally incorrect answers that general-purpose chatbots can produce. The RAG Fusion and LLM Generation stage then re-ranks the retrieved context and drafts a grounded, source-cited answer, which is back-translated into the farmer's own language before being delivered through the same channel — as synthesised speech, app text, or a WhatsApp reply.

8. System Requirements

a. Hardware Requirements

1. Farmer-side device: smartphone (for the app or WhatsApp) or any basic phone (for IVR/voice calls)

2. Server-side compute for model inference (cloud GPU/CPU instance)

3. Microphone/speaker for voice input and output (device-side)

4. Stable internet connectivity for live API calls and cloud inference
5. IVR telephony gateway for voice-call access

b. Software Requirements

1. Python programming language
2. Speech-to-text (ASR) and text-to-speech (TTS) engines for Indian regional languages
3. Machine translation model (regional language ↔ English)
4. Large language model (LLM) for grounded answer generation
5. Vector database for retrieval-augmented generation (RAG)
6. Backend web framework (e.g., FastAPI/Django) and REST APIs
7. Third-party APIs: weather forecast, mandi price, government scheme databases
8. WhatsApp Business API / IVR integration platform
9. Cloud hosting and deployment environment

9. Methodology / Implementation Plan

Development Methodology: Incremental approach

Implementation Steps:

1. Requirement gathering and study of regional agricultural information needs
2. Collecting and verifying agricultural advisory documents to build the knowledge base
3. Building the language identification, ASR, and translation pipeline
4. Building the intent classifier for the five query categories
5. Implementing the retrieval layer (vector search + live API integration)
6. Implementing the RAG fusion, trust/validation, and answer-generation pipeline
7. Implementing back-translation and multi-channel delivery (app, WhatsApp, IVR)
8. Testing with sample farmer queries and refining accuracy
9. Pilot testing and feedback collection from farmers

10. Expected Outcome

The expected outcome is a functional prototype of Bhunaga capable of accepting a farmer's voice or text query in their own language, correctly identifying its intent, retrieving relevant and trustworthy information, and returning an accurate, localised, source-grounded answer through the farmer's preferred channel (app, WhatsApp, or IVR call).

11. Applications / Use Cases

1. Sowing and crop-selection guidance for smallholder farmers
2. Pest identification and spraying/dosage advisory
3. Localised weather forecasts for farm planning
4. Real-time mandi (market) price lookup before selling produce
5. Guidance on government agricultural schemes, loans and subsidies
6. Marketplace/selling assistance connecting farmers with buyers

12. Future Enhancements / Future Scope

1. Expanding language coverage to additional regional and dialectal variations
2. Offline or low-bandwidth modes for areas with poor connectivity
3. Personalised advisory based on a farmer's location, crop history and land records
4. Integration with on-field IoT sensors and satellite/drone crop-health data
5. Support for in-app marketplace transactions and buyer-seller matching
6. Continuous learning from farmer feedback to improve answer accuracy over time

13. References

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